

Paper II — SoK

SoK: The Quantifier Gap Between Certified Robustness and Airworthiness Assurance

Integrated proposal, broadened from P3 · RobustUAVs.ai · Roger Nick Anaedevha · 2026-10-01
Target: IEEE Symposium on Security and Privacy, SoK track · fallback: DASC or ACM Computing Surveys

Abstract

Certified-robustness research and aviation assurance both promise guarantees about learned systems, and both are now being asked to meet: EASA’s learning-assurance framework and the forthcoming ED-324 expect evidence about machine-learning constituents, and the robustness literature produces certificates in quantity. We argue that the two rarely meet because they quantify over different things. A certified radius is a statement about *every perturbation of one input*; a contingency volume is a statement about *one operation*; a safety objective per flight hour is a statement about *a population of operations*. Evidence transfers from a guarantee to an objective only when the guarantee’s quantifier dominates the objective’s, or when a *lifting argument* — a dynamics bound, a concentration inequality, a runtime monitor — bridges them, at the price of an assumption that itself needs evidence. We propose to systematise the certified-guarantee literature for learned perception, navigation and control by quantifier, to code SORA and EASA learning-assurance objectives the same way, and to tabulate which transfers are sound, which require a lift, and which are category errors. A SORA Annex A evidence generator demonstrates the sound path end to end.

1 The observation

P3 proposed mapping one certificate — the parent manuscript’s [1] — onto SORA and ED-324 vocabulary, and its own honesty note predicted the likely finding: *certified radii are per-input, while assurance objectives are per-operation*. That sentence is more general than P3 used it for. It is a claim about an entire literature, and it is testable.

It is also incomplete in one direction, which is what makes it worth a systematisation rather than a remark. Aviation does not quantify only per operation. A SORA contingency volume is sized for one operation; but a safety objective expressed per flight hour, and EASA’s learning-assurance expectation that a model generalises across its Operational Design Domain, are statements about *populations*. So the two communities do not simply sit at two ends of one axis. Each produces statements at several quantifier levels, and the question is which pairs line up.

2 The framework: four quantifier levels

Level	Quantifies over	Typical guarantee	Typical assurance objective
Q_{in}	all perturbations of one input	certified radius: randomized smoothing [3], bound propagation [13], SMT [10]	rarely stated at this level
Q_{tr}	all disturbances along one trajectory, over a horizon	reachability tube; Lipschitz–Grönwall [7]	position-holding error; flight technical error
Q_{op}	one complete operation	mission predicate (spatial <i>and</i> temporal); runtime assurance with handover [12]	SORA contingency volume [8]
Q_{pop}	a distribution of operations	PAC-Bayes [11]; regret [2]; mission completion rate	per-flight-hour safety objectives; generalisation over the ODD [6]

Definition 1 (Sound transfer). *A guarantee at level Q_g discharges an objective at level Q_o only if Q_g quantifies over a superset of what Q_o requires and under a perturbation model that covers the objective’s hazard. Otherwise it is at most evidence, or it is a category error.*

Definition 2 (Lift). A lift is an argument that carries a guarantee from one level to another — $Q_{in} \rightarrow Q_{tr}$ by a dynamics bound, $Q_{tr} \rightarrow Q_{op}$ by a horizon and re-anchoring argument, $Q_{op} \rightarrow Q_{pop}$ by a concentration or PAC argument. Every lift introduces an assumption, and that assumption is itself an assurance obligation.

The parent manuscript’s composition chain is a worked instance: Q_{in} (a perturbation bound on the navigation input) lifted to Q_{tr} by Grönwall under a Lipschitz hypothesis, to Q_{op} by a corridor predicate under re-anchoring, and to Q_{pop} by a mission distribution. Each of its three stated conditions is exactly one lift’s assumption. That is why it serves as the SoK’s running example rather than its subject.

3 Hypotheses the systematisation tests

These are predictions, and the paper is only worth writing if any of them can come out false.

- H1 Most certified-robustness work for learned perception sits at Q_{in} , and therefore discharges no assurance objective without a lift.** The corpus will show the proportion.
- H2 Lifts exist in the literature but are rarely accounted for as obligations.** Papers that lift a guarantee seldom list the lift’s assumption as something requiring evidence. The coding scheme records this per paper.
- H3 The Q_{pop} objectives — per flight hour, generalisation over the ODD — are the natural match for statistical guarantees, yet statistical guarantees for learned perception are the least developed class.** If true, this is the research agenda the SoK hands the community.
- H4 Runtime assurance is the dominant practical bridge** for objectives no offline certificate discharges, because it converts a per-operation obligation into a per-input check plus a verified fallback.

■ *H1 could fail. If the corpus turns out to contain substantially more Q_{tr} and Q_{pop} work than expected — closed-loop reachability for neural controllers is an active area — the gap is narrower than this proposal claims, and the paper must report that. An SoK whose finding is “it mostly lines up” would be less exciting and still publishable, but only if it says so rather than choosing a corpus that confirms the thesis.*

4 Method

4.1 The guarantee corpus

- **Sources.** IEEE S&P, ACM CCS, USENIX Security, NDSS; NeurIPS, ICML, ICLR; CAV, HSCC, ICCPS, EMSOFT; DASC and the SAE/AIAA avionics venues. Years 2017–2026.
- **Inclusion.** A formal or statistical guarantee on a learned component used, or evaluated, for perception, state estimation, navigation or control of an autonomous vehicle. Benchmarks without a guarantee are out; guarantees on non-learned components are out.
- **Search.** Keyword search seeded from the canonical systems ([10, 13, 3, 9]), then forward and backward citation snowballing until saturation.
- **Expected size.** Of the order of 80–150 papers. This is a prediction, not a count: **the corpus has not been collected.**

4.2 The objective corpus

SORA 2.5 operational safety objectives and the contingency-volume definition in Annex A [8]; the EASA AI Concept Paper Issue 2 learning-assurance objectives [6]; the MLEAP findings [5]; and ED-324 / ARP6983 [4] at the level of concepts only, since it is in preparation.

4.3 Coding scheme

Each guarantee and each objective is coded on the same fields, so they can be joined:

Field	Values
Quantifier level	Q_{in} , Q_{tr} , Q_{op} , Q_{pop}
Perturbation model	norm-bounded input; sensor delay; sensor bias; adversarial spoofing; distribution shift; none
Lifts used	dynamics, horizon/re-anchoring, concentration, runtime monitor; each with its stated assumption
Assumption accounted for?	yes / stated but not evidenced / unstated
Evidence class	proof; measured; simulated; modelled
Domain	perception, estimation, navigation, control, end-to-end

Two coders code independently; disagreements are resolved by discussion, and Cohen's κ is reported for the quantifier and assumption fields, which are the ones the findings rest on.

4.4 The transfer table

The output joins the two corpora on quantifier level and perturbation model and classifies every (guarantee class, objective) pair as **discharge**, **evidence via lift** (naming the lift), or **category error**. P3's discharge / evidence / untouched gap analysis becomes one slice of this table.

5 The evidence generator

Carried over from P3, and now with a sharper purpose: it demonstrates one *sound* transfer path end to end. From a certified run of the parent project it emits a SORA Annex A-shaped evidence pack in which

- the Operational Design Domain is the parent's *measured* operating region (attitudes off singularity, airspeed below stall, actuators unsaturated), which P3 noticed is an ODD in all but name;
- the contingency extent is the certified tube radius;
- each lift's assumption appears as an explicit, open obligation rather than disappearing into the result; and
- every figure carries its provenance tag.

The third point is the generator's real contribution. A compliance pack that lists the certificate but hides its assumptions would be precisely the category error the SoK exists to name.

6 What the SoK delivers

- **A quantifier framework** that places certified guarantees and assurance objectives on one axis.
- **A coded corpus** of certified guarantees for learned autonomy, released with the coding scheme and agreement statistics.
- **The transfer table**: which guarantees discharge which objectives, which need which lift, and which pairings are category errors.
- **An accounting of lifts as obligations**, testing whether the literature treats them as such.
- **A research agenda**: the empty cells of the transfer table, ranked by how many objectives they would discharge.
- **An evidence generator** demonstrating one sound path.

7 Relation to the parent manuscript

P3 already said that it closes none of the parent’s limitations, and the SoK does not change that. What it changes is how one of them reads. The parent’s guarantee bounds a population of missions rather than a single flight, which a reviewer can read as a weakness. The SoK shows that this is a property of every $\mathbf{Q}_{pop}$ guarantee and of every per-flight-hour objective, so it is the correct level for an airworthiness argument rather than a compromise. That is a reframing, and it should be presented as one, not as a closure.

8 Risks

- **A thin or biased corpus.** The main way an SoK fails at S&P. Stated inclusion criteria, snowballing to saturation, two coders and a reported κ are the defence. If time forces a smaller corpus, the paper says how much smaller and drops the claims that need the full one.
- **The standards move.** ED-324 is in preparation. Key every claim to a stable concept — ODD, contingency volume, learning assurance — and confine clause numbers to an appendix.
- **Regulatory over-claim.** Nothing here certifies an aircraft. The paper maps mathematical guarantees onto an evidence framework, and the abstract must already say so.
- **Being read as a survey.** The quantifier framework and the transfer table are what distinguish an SoK from a survey. If either is weak, the fallback is ACM Computing Surveys, and the paper should be retargeted rather than oversold.

9 Page plan and timeline

Section	pp	Section	pp
Introduction	1.25	The transfer table	2.0
The quantifier framework	1.75	Lifts as obligations	1.25
Method and corpus	1.5	Evidence generator (case)	1.0
Guarantees by level	1.75	Research agenda	1.0
Objectives by level	1.0	Related SoKs, conclusion	0.5
Body total			13.0

About three months, and it needs no measurement, so it can start at once and run alongside Paper I’s hardware campaign. Weeks 1–6: search and coding. Weeks 5–8: transfer table and generator. Weeks 9–12: writing. Submit it before Paper I, so that Paper I can cite its vocabulary.

References

- [1] Anonymous. From Wire to Flight: A Conditional Mission-Completion Guarantee for UAVs. under submission, 2026. the parent manuscript; anonymised for review.
- [2] Sanjeev Arora, Elad Hazan, and Satyen Kale. The Multiplicative Weights Update Method: A Meta-Algorithm and Applications. *Theory of Computing*, 8(1):121–164, 2012.
- [3] Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Randomized Smoothing. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2019.
- [4] EUROCAE WG-114 and SAE G-34. ED-324 / ARP6983: Process Standard for Development and Certification/Approval of Aeronautical Products Implementing AI. Technical report, EUROCAE / SAE International, 2026. in preparation.
- [5] European Union Aviation Safety Agency. Machine Learning Application Approval (MLEAP). EASA research project. accessed 2026-08-13.
- [6] European Union Aviation Safety Agency. Artificial Intelligence Concept Paper Issue 2: Guidance for Level 1 and 2 Machine Learning Applications, 2024. accessed 2026-08-13.

- [7] Thomas H. Grönwall. Note on the Derivatives with Respect to a Parameter of the Solutions of a System of Differential Equations. *Annals of Mathematics*, 20(4):292–296, 1919.
- [8] Joint Authorities for Rulemaking of Unmanned Systems. JARUS guidelines on Specific Operations Risk Assessment (SORA), edition 2.5 — annex A. Technical Report JAR-DEL-WG6-D.04, JARUS, 2024.
- [9] Kyle D. Julian, Jessica Lopez, Jeffrey S. Brush, Michael P. Owen, and Mykel J. Kochenderfer. Policy Compression for Aircraft Collision Avoidance Systems. In *Proc. IEEE/AIAA Digital Avionics Systems Conference (DASC)*, 2016.
- [10] Guy Katz, Clark Barrett, David L. Dill, Kyle Julian, and Mykel J. Kochenderfer. Reluplex: An Efficient SMT Solver for Verifying Deep Neural Networks. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, 2017.
- [11] David A. McAllester. Some PAC-Bayesian Theorems. In *Proc. Conf. on Computational Learning Theory (COLT)*, pages 230–234, 1998.
- [12] Lui Sha. Using Simplicity to Control Complexity. *IEEE Software*, 18(4):20–28, 2001.
- [13] Shiqi Wang, Huan Zhang, Kaidi Xu, Xue Lin, Suman Jana, Cho-Jui Hsieh, and J. Zico Kolter. Beta-CROWN: Efficient Bound Propagation with Per-neuron Split Constraints for Neural Network Robustness Verification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.